MATH 3115 MSc & SED COMPLEX ANALYSIS

Lecture 10: ANALYTIC FUNCTIONS

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Lesson Outcome

By the end of this lesson you should be able to:

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INTRODUCTION

- Here, we study various elementary functions that are encountered in calculus.
- We will show how to arrive at natural definitions for the complex functions corresponding to the exponential, logarithmic , trigonometric, and hyperbolic functions.
- These functions will be extensions of the real variable functions and will be consistent with the definitions of the complex functions.

EXPONENTIAL FUNCTION

Definition 1

The exponential function is defined as $f(x) = e^x, x \in \mathbb{R}$, and e is the base called a natural base.

The exponential function preserves the laws of exponents:

(a) $e^x e^y = e^{x+y}$
(b) $e^0 = 1$
(c) $\frac{d}{dx}(e^{ax}) = ae^{ax}$

Exponential functions

Definition 2

If $z = x + iy$ for $x, y \in \mathbb{R}$, then we have
$e^z = e^{x+iy} = e^x e^{iy}$

- Define $f(y) = e^{iy}$. Then $f(0) = e^{i(0)} = e^0 = 1$
  $\frac{df}{dy} = ie^{iy} = if(y)$

Exponential functions

Definition 3

If $z = x + iy$, then $e^z = e^{x+iy}$ is defined to be the complex number $e^z = e^{x+iy} = e^x e^{iy} = e^x(\cos y + i \sin y)$. Hence $e^z = e^x(\cos y + i \sin y)$.

- when z is real, we have $y = 0, e^z = e^{x+iy} = e^{x+0} = e^x$
- When $z = 0$, we have $e^z = 1$, and $\arg e^z = y$

Theorem 1

If $x_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$ then $e^{z_1}e^{z_2} = e^{z_1+z_2}$

Proof

$$
\begin{aligned}
e^{z_1}e^{z_2} &= e^{x_1+iy_1}e^{x_2+iy_2} \\
&= e^{x_1}(\cos y_1 + i\sin y_1)\, e^{x_2}(\cos y_2 + i\sin y_2) \\
&= e^{x_1+x_2}\left[(\cos y_1 \cos y_2 - \sin y_1 \sin y_2) + i(\sin y_1 \cos y_2 + \sin y_2 \cos y_1)\right]. \\
&= e^{x_1+x_2}(\cos(y_1+y_2) + i\sin(y_1+y_2)) \\
&= e^{x_1+x_2}e^{i(y_1+y_2)} \\
&= e^{x_1+x_2+iy_1+iy_2} \\
&= e^{z_1+z_2}
\end{aligned}
$$

EXPONENTIAL FUNCTIONS CONT'D

Theorem 2

(a) $\forall z \in \mathbb{C}$, we have $e^z \neq 0$
(b) If $z = x + iy$, we have $\left|e^{iy}\right| = 1$ and $\left|e^z\right| = e^x$
(c) $e^z = 1$ if $z = 2\pi ki, k \in \mathbb{Z}$
(d) If $e^{z_1} = e^{z_2}$, then $z_1 - z_2 = 2k\pi i, k \in \mathbb{Z}$

Proof of (a)

If $z_1 = z$ and $z_2 = -z$, then $e^{z_1} \cdot e^{z_2} = e^{z_1 + z_2}$

$$= e^{z-z} = e^0 = 1$$

since the product is never zero, the factor can never be zero. Therefore $e^z \neq 0$.

EXPONENTIAL FUNCTIONS CONT'D

Proof of (b)

$$\left|e^{iy}\right| = \mid \cos y + i \sin y \mid$$
$$= \sqrt{\cos^2 y + \sin^2 y} = \sqrt{1} = 1.$$
$$\text{Also } \left|e^{z}\right| = \left|e^{x+iy}\right|$$
$$= \left|e^{x} e^{iy}\right|$$
$$= \left|e^{x}\right| \left|e^{i} y\right|$$
$$= \left|e^{x}\right|$$
$$= e^{x}$$

- Proofs for (c) and (d) is left for the reader as an exercise

EXPONENTIAL FUNCTIONS CONT'D

Definition 4

The function $f(z)$ is said to be periodic of period λ if for all z in th Domain of f, we have $f(z) = f(z + \lambda)$

Example 1

e^z is of period $2\pi i$ because

$$\begin{aligned} e^{z+2\pi i} = e^z e^{2\pi i} &= e^z(\cos 2\pi + i \sin 2\pi) \\ &= e^z(1+0) \\ &= e^z \end{aligned}$$

EXPONENTIAL FUNCTION CONT'D

Theorem 3

The exponential function $f(z) = e^z$ is analytic for all values of z. Moreover $\frac{d}{dz}(e^z) = e^z$

Proof.

The proof is left for the reader as an exercise $\square$

Example 2

Show that $e^{\bar{z}} = \overline{e^z}$, and $e^{\bar{z}}$ is not an analytic function of z in any domain D.

Solution to Example 2

$$e^{\bar{z}} = e^{\overline{x+iy}} = e^{x-iy} = e^{x}e^{-iy} = e^{x}\left(\cos(-iy) + i\sin(-iy)\right)'$$
$$= e^{x}(\cos y - i\sin y)$$

Similarly $\overline{e^{z}} = \overline{e^{x+iy}} = e^{\overline{x(\cos y + i\sin y)}}$

$$= e^{\bar{x}}(\cos y + i\sin y)$$

$$= e^{x}(\cos y - i\sin y)$$

So $e^{\bar{z}} = \overline{e^{z}}$

To show that $\bar{z}$ is not analytic function of z in any domain D, we let $u + iv = e^{\bar{z}} = e^{x}(\cos y - i\sin y)$ so that $u = e^{x}\cos y$ and $V = -e^{x}\sin y_1$ Computing the partial derivatives, we get $u_x = +e^{x}\cos y, \quad u_y = -e^{x}\sin y, v_x = -e^{x}\sin y, \quad v_y = -e^{x}\cos y$.

We note that $u_x = e^{x}\cos y \neq v_y = -e^{x}\cos yv$ and $V_x = -e^{x}\sin y \neq -v_y = e^{x}\sin y$. Hence Cauchy Riemann Equations are not satisfied. Then we have $f(z) = e^{z}$ is not analytic for $z \in D$.

MODULUS AND ARGUMENT OF e^z

INTRODUCTION

- We know that $e^z = e^{x+1y} = e^x(\cos y + i \sin y)$
- This means that $r = e^x$ and $\theta = y$
- Hence the modulus of e^z is e^x and the argument of e^z is y.in That is

$$|e^z| = \left|e^{x+iy}\right| = \left|e^x e^{iy}\right| = |e^x| \left|e^{iy}\right| = e^x$$
$$\arg\left(e^z\right) = \arg\left(e^x e^{iy}\right) = y + 2\pi n, n \in \mathbb{Z}$$

- Consider $w = e^z$. Using polar coordinates $w = r(\cos\theta + i \sin\theta), r = |w|$ and $\theta = \arg w$ since r is a positive real number, then $\ln r$ exists and $w = e^{\ln r}(\cos\theta + i \sin\theta) = e^{(\ln r + (\theta)}$
- Therefore we have $z = \ln|w| + i(\operatorname{Arg} w + 2\pi n), n \in \mathbb{Z}$

Example 3

Find all the values of z for which the equation $e^z = \sqrt{3} - i = w$ holds true.

Solution

Consider $\sqrt{3} - i$, $\quad r = \sqrt{(\sqrt{3})^2 + (-1)^2} = \sqrt{3+1} = \sqrt{4} = 2$

$$\theta = \tan^{-1}(-1/\sqrt{3}) = -\pi/6$$

$\sqrt{3} - i = 2\left(\cos\frac{\pi}{6} - i\sin\pi/6\right)$. Now we see that

$$r = |w| = 2$$

$$\theta = -\pi/6 + 2\pi n, \quad n \in \mathbb{Z}$$

so $z = \ln 2 + i(-\pi/6 + 2\pi n), n \in \mathbb{Z}$

TRIGONOMETRIC FUNCTIONS

TRIG FUNCTION IN COMPLEX PLANE

- We have seen that $e^{iy} = \cos y + i \sin y$ and $e^{-iy} = \cos y - i \sin y$.
- Upon addling the two expressions/equations, we have

$$\begin{cases} + & e^{iy} = \cos y + i \sin y \\ & e^{-iy} = \cos y - i \sin y \end{cases} \Rightarrow e^{iy} + e^{-iy} = 2\cos y$$

$$\cos y = \frac{e^{iy} + e^{-iy}}{2}$$

- In a similar way, $\sin y = \frac{e^{iy} - e^{-iy}}{2i}$
- These trigonometric functions are extended to the domain of a complex variable.

Definition 5

Given a complex number z, we define

$$\sin z = \frac{e^{iz} - e^{-iz}}{2i}, \quad \cos z = \frac{e^{iz} + e^{-iz}}{2}$$

- Note that $\sin z$ and $\cos z$ are both periodic with period $2\pi i$.

Theorem 4

The functions $\sin z$ and $\cos z$ are analytic for all values of z. Moreover,

$$\frac{d}{dz}(\sin z) = \cos z, \quad \frac{d}{dz}(\cos z) = -\sin z$$

Definition 6

A point z_0 for which $f(z_0) = 0$ is called a zero of the function $f(z)$.

Theorem 5

The zeros of the functions $\sin z$ and $\cos z$ are given respectively by

$$z = n\pi \quad \text{and} \quad Z = \frac{\pi}{2} + n\pi, \quad n \in \mathbb{Z}$$

THE END